



ITSimplera Institute

Week 1

Data Analysis & Business Understanding

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1 Introduction

Exploratory Data Analysis (EDA) is an important step in the data analysis process that helps understand the characteristics, quality, and patterns of a dataset before applying machine learning algorithms. It enables analysts to identify missing values, duplicate records, outliers, trends, and relationships among variables.

In this project, an Exploratory Data Analysis was performed on the Online Retail II dataset. The dataset contains transactional records of a UK-based online retail company from 2009 to 2011. The objective was to analyze customer purchasing behavior, identify the best-selling products, evaluate sales across different countries, examine monthly revenue trends, and detect data quality issues without modifying the dataset.

2 Objectives

- To load and understand the Online Retail II dataset.
- To identify missing values and duplicate records.
- To analyze the top-selling products based on quantity and revenue.
- To evaluate sales performance across different countries.
- To analyze monthly revenue trends.
- To study relationships among numerical variables using a correlation heatmap.
- To detect outliers using box plots.
- To derive meaningful business insights from the analysis.

3 Dataset Description

Dataset Name: Online Retail II Dataset

Source: UCI Machine Learning Repository

The dataset contains transaction records from an online retailer operating in the United Kingdom between 2009 and 2011. The dataset contains the following attributes:

Column	Description
Invoice	Invoice number
StockCode	Product code
Description	Product description
Quantity	Quantity purchased
InvoiceDate	Date and time of transaction
Price	Unit price of the product

Column	Description
Customer ID	Unique customer identifier
Country	Customer's country

500,000+
TRANSACTION RECORDS

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-12-01 07:45:00	6.95	13085.0	United Kingdom
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13085.0	United Kingdom
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13085.0	United Kingdom
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-12-01 07:45:00	2.10	13085.0	United Kingdom
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-12-01 07:45:00	1.25	13085.0	United Kingdom

Figure 3.1 — Snapshot of dataset attributes

4 Methodology

The following steps were performed during the analysis:

- 01 Data Loading**
 The dataset was loaded into Google Colab using the Pandas library. The first few rows of the dataset were displayed to verify successful loading.
- 02 Basic Information**
 Basic information about the dataset was obtained, including the number of rows and columns, column names, data types, and sample records — helping establish the structure of the dataset.

Shape of the dataset: (525461, 8)

Column Names:

```
Index(['Invoice', 'StockCode', 'Description', 'Quantity', 'InvoiceDate',
      'Price', 'Customer ID', 'Country'],
      dtype='object')
```

Data Types:

```
Invoice          object
StockCode        object
Description       object
Quantity         int64
InvoiceDate      datetime64[ns]
Price            float64
Customer ID      float64
Country          object
dtype: object
```

Figure 4.1 — Dataset structure overview

03 Missing Value Analysis

Missing values were identified for every column. Description contained a small number of missing values, Customer ID contained a significant number of missing values, and all other columns were complete. No missing values were removed, as the assignment required observation only.

04 Duplicate Row Analysis

Duplicate rows were counted to identify repeated transactions. The dataset contained 6,865 duplicate rows, indicating potential data quality issues. No duplicates were removed during this stage.

05 Product Analysis

Product performance was analyzed in two ways — top 10 products by quantity sold, and top 10 products by revenue generated. Bar charts were created to visualize the results, helping identify products with the highest customer demand and those contributing the most revenue.

06 Country-wise Sales Analysis

Revenue generated by each country was calculated and visualized with a bar chart. The analysis identified the United Kingdom as the highest revenue-generating country.

07 Monthly Revenue Trend

Revenue was grouped by month using the InvoiceDate column. A line chart was created to observe revenue changes over time, showing seasonal variations in customer purchases.

08 Correlation Analysis

A correlation matrix was calculated using the numerical columns Quantity, Price, and Revenue. A heatmap was created using Seaborn to visualize relationships among these variables.

09 Outlier Detection

Box plots were created for Quantity, Price, and Revenue. The box plots revealed several extreme values within the dataset. As required, no outliers were removed during the analysis.

5 Results

Missing Values



Missing Values in Each Column:

```
Invoice      0
StockCode    0
Description  2928
Quantity     0
InvoiceDate  0
Price        0
Customer ID  107927
Country      0
```

dtype: int64

Percentage of Missing Values:

```
Invoice      0.000000
StockCode    0.000000
Description  0.557225
Quantity     0.000000
InvoiceDate  0.000000
Price        0.000000
Customer ID  20.539488
Country      0.000000
```

dtype: float64

Figure 5.1 — Missing values by column

These missing values indicate that not every transaction can be associated with a customer.

Duplicate Rows

Number of Duplicate Rows: 6865

Figure 5.2 — Duplicate row count

A total of 6,865 duplicate rows were identified. These duplicate records should be addressed during data preprocessing before model development.

Top Selling Products

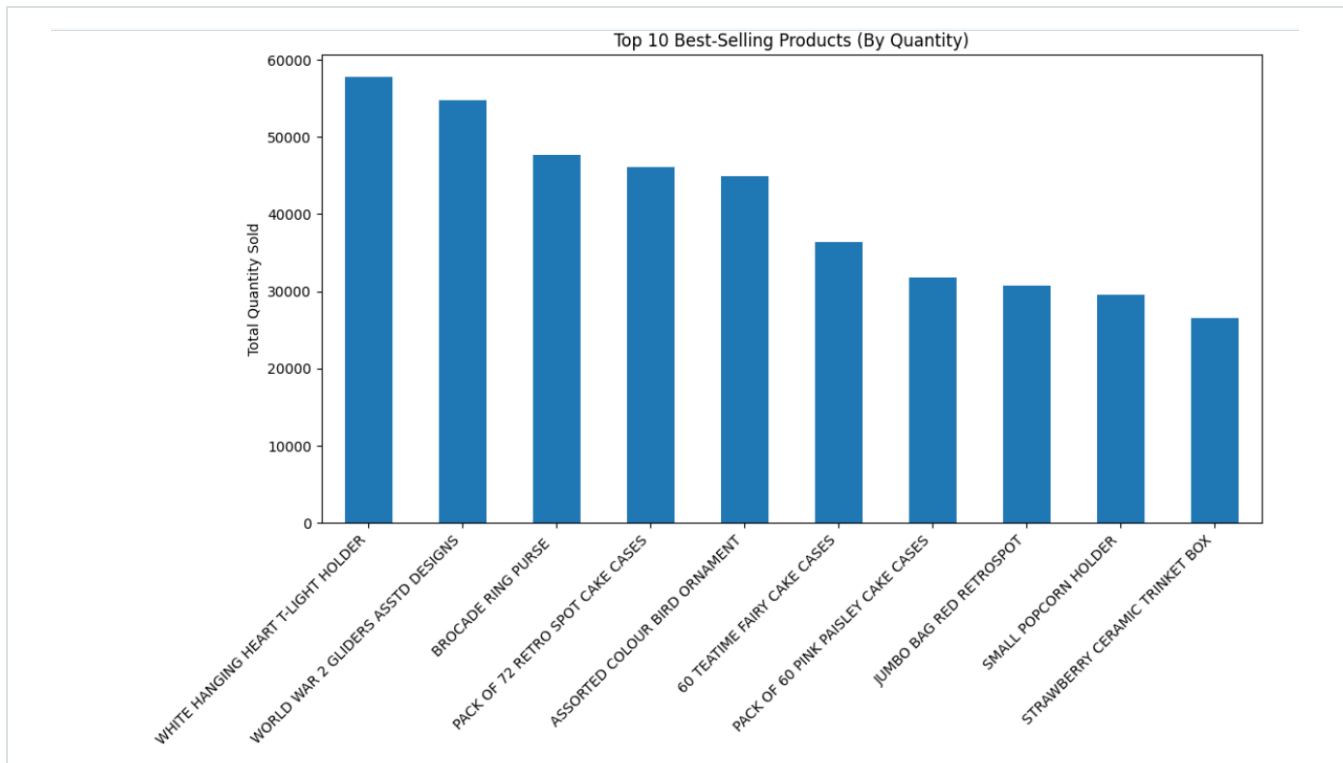


Figure 5.3 — Top 10 products by quantity and revenue

The product analysis showed that several products consistently appeared among the highest-selling products by quantity and revenue, including:

- WHITE HANGING HEART T-LIGHT HOLDER
- REGENCY CAKESTAND 3 TIER
- ASSORTED COLOUR BIRD ORNAMENT

The analysis also showed that products generating the highest revenue are not always those sold in the greatest quantities.

Country-wise Sales

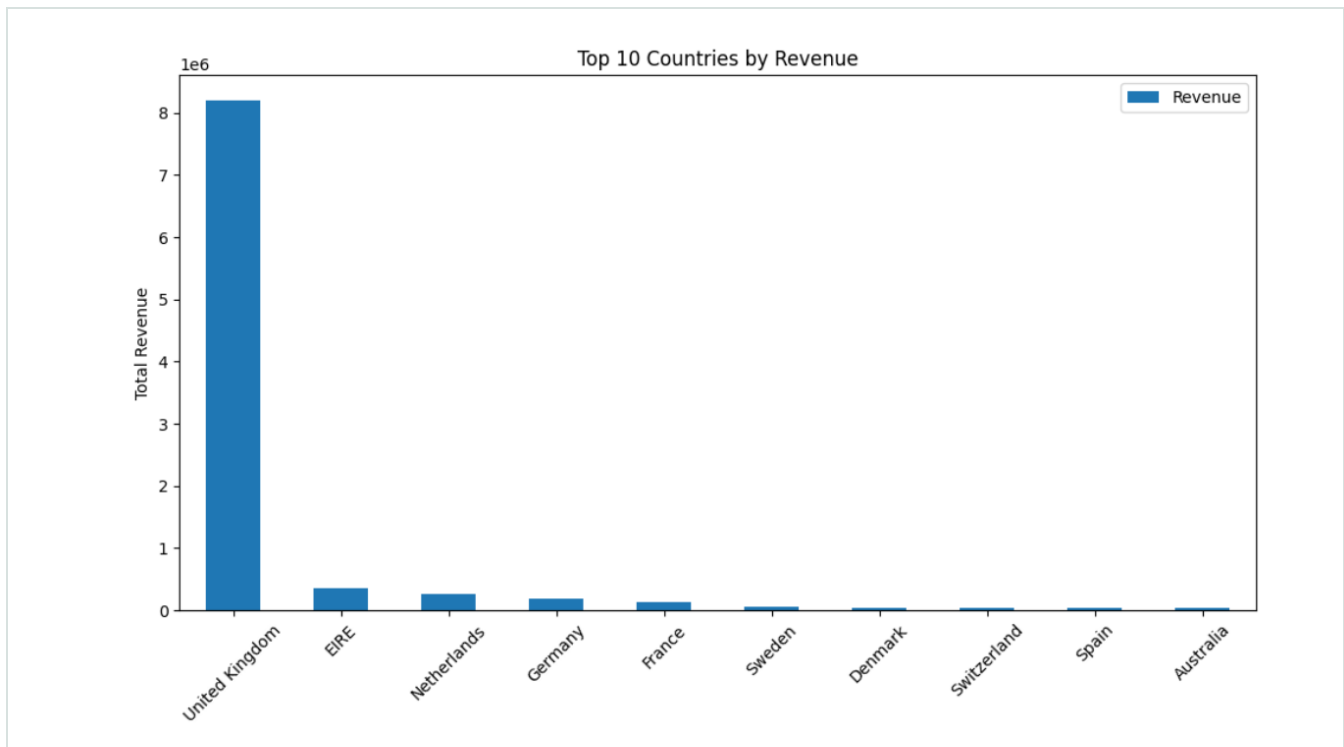


Figure 5.4 — Revenue by country

The United Kingdom generated the highest overall revenue, indicating that the retailer's primary customer base is located within the United Kingdom.

Monthly Revenue Trend

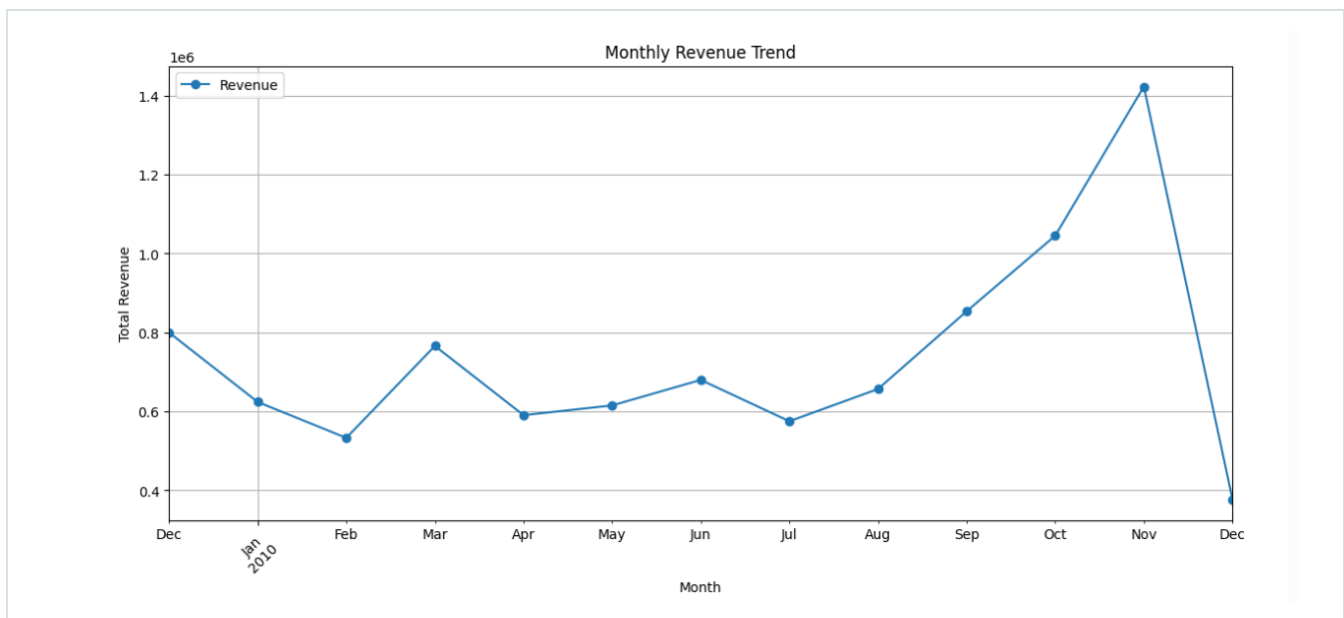


Figure 5.5 — Monthly revenue trend

The monthly revenue graph showed fluctuations throughout the year. The highest revenue was observed during November, suggesting increased customer purchases during the holiday shopping season.

Correlation Analysis

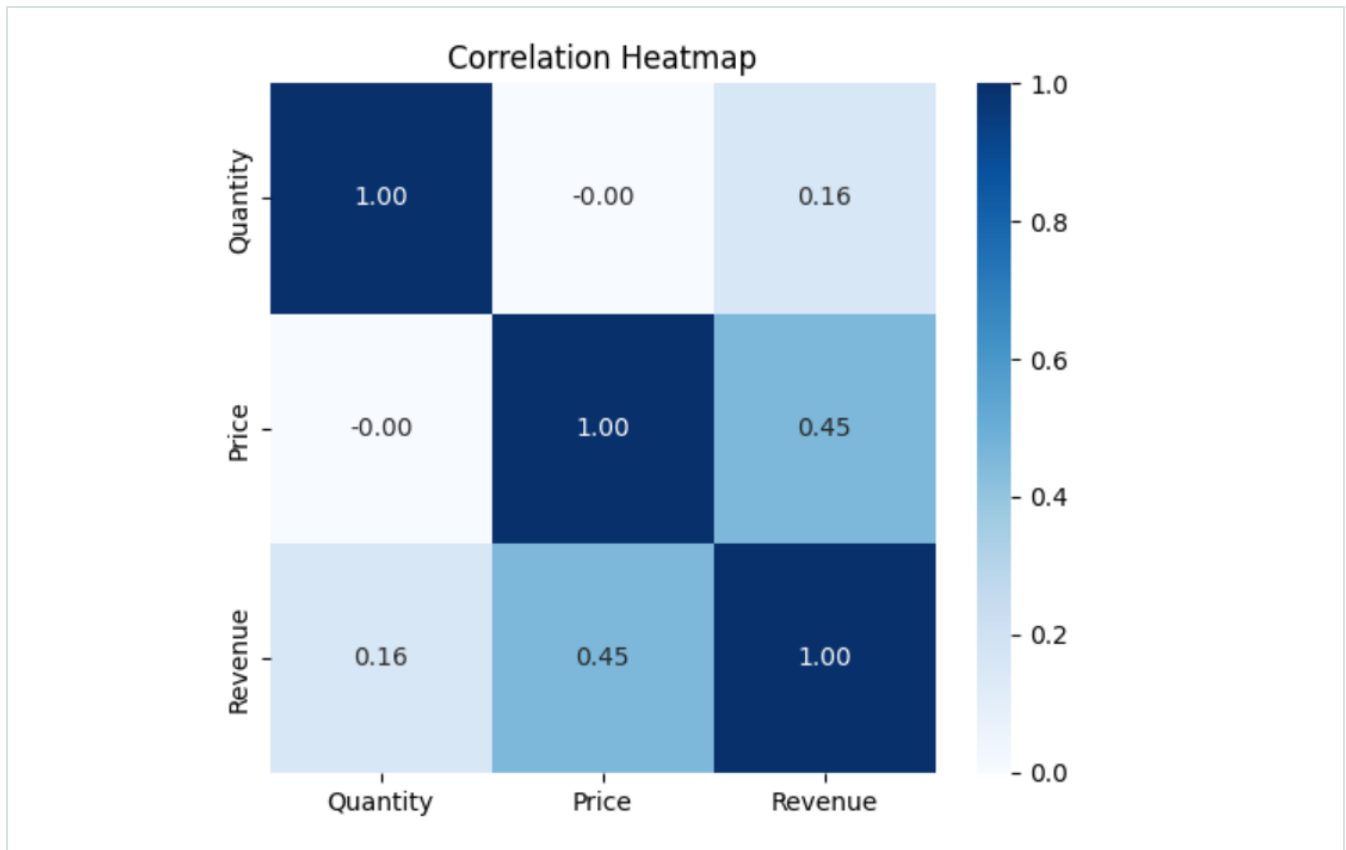


Figure 5.6 — Correlation heatmap

The heatmap produced the following observations:

- Quantity and Revenue showed a weak positive relationship.
- Price and Revenue showed a moderate positive relationship.
- Quantity and Price showed almost no correlation.

This indicates that revenue depends more on product pricing than simply selling higher quantities.

Outlier Detection

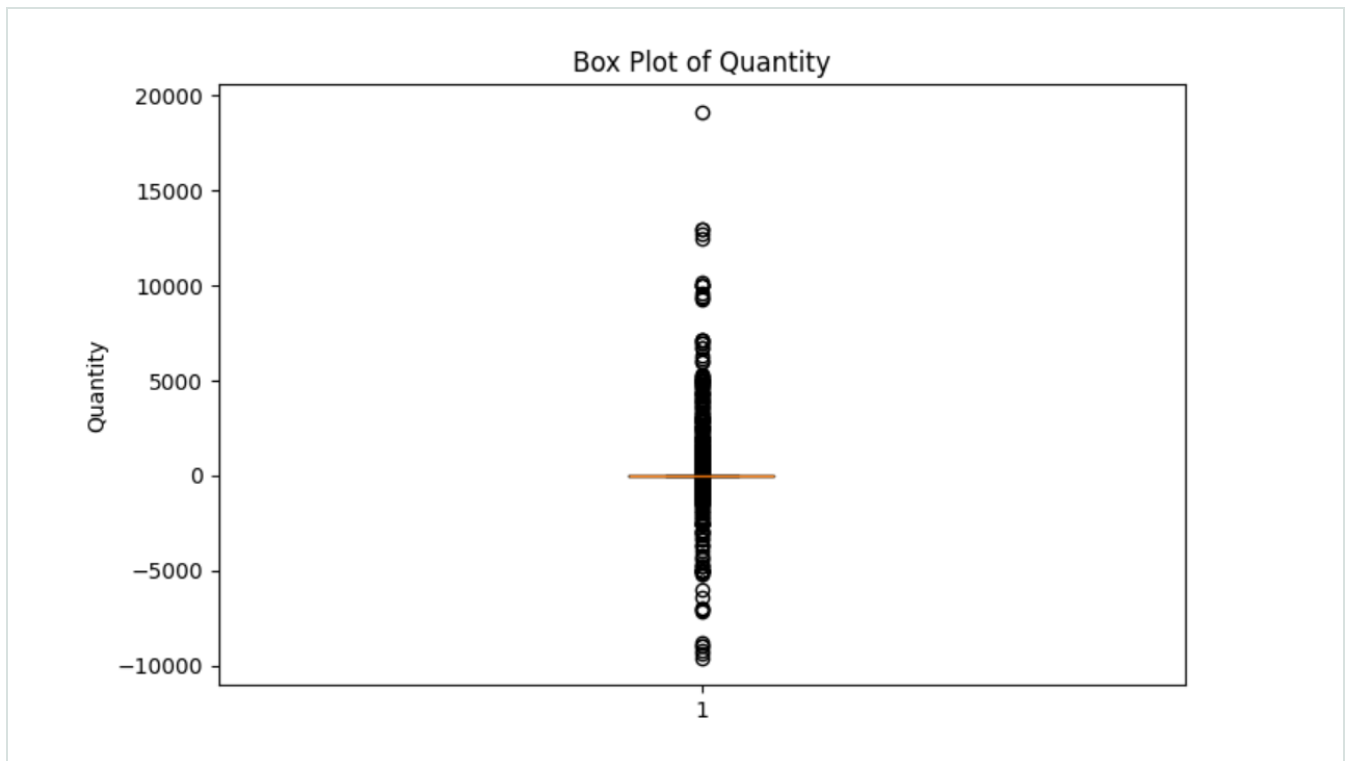


Figure 5.7 — Box plot: Quantity

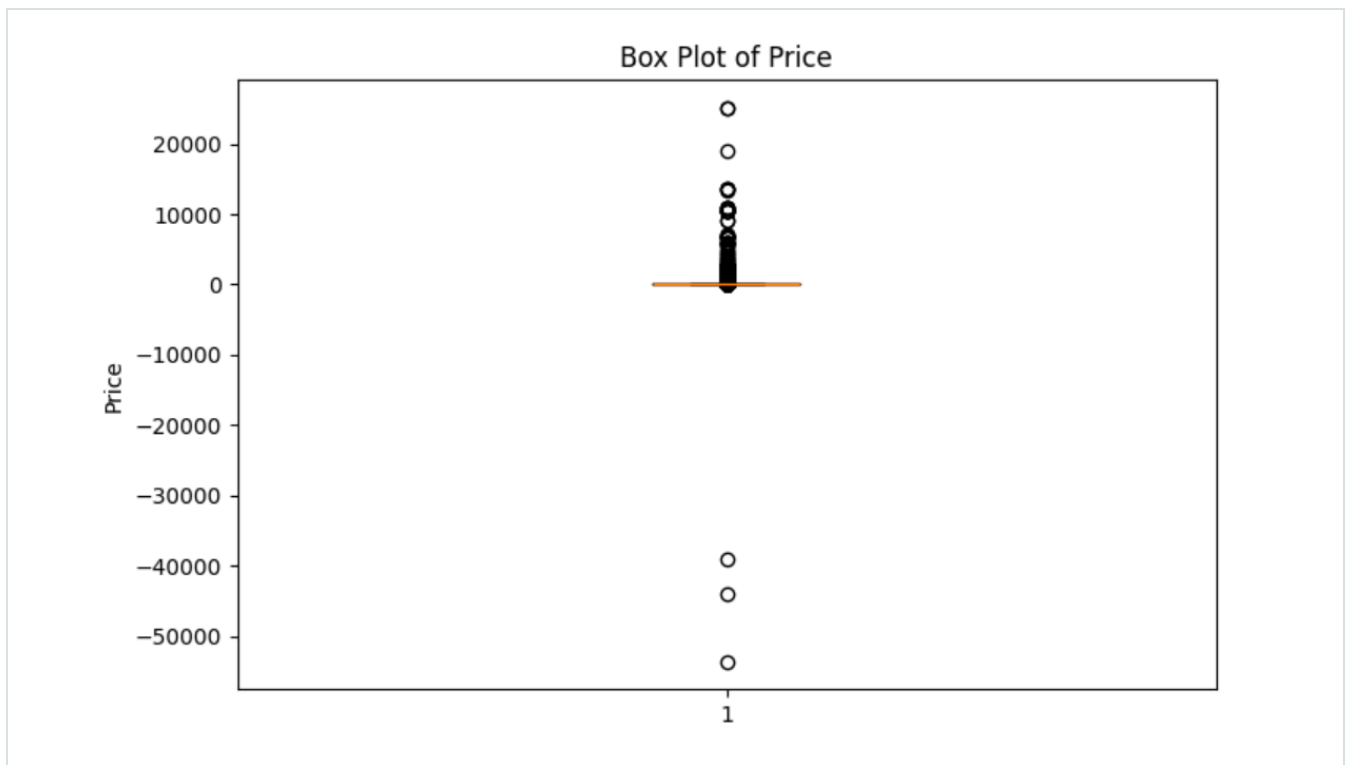


Figure 5.8 — Box plot: Price

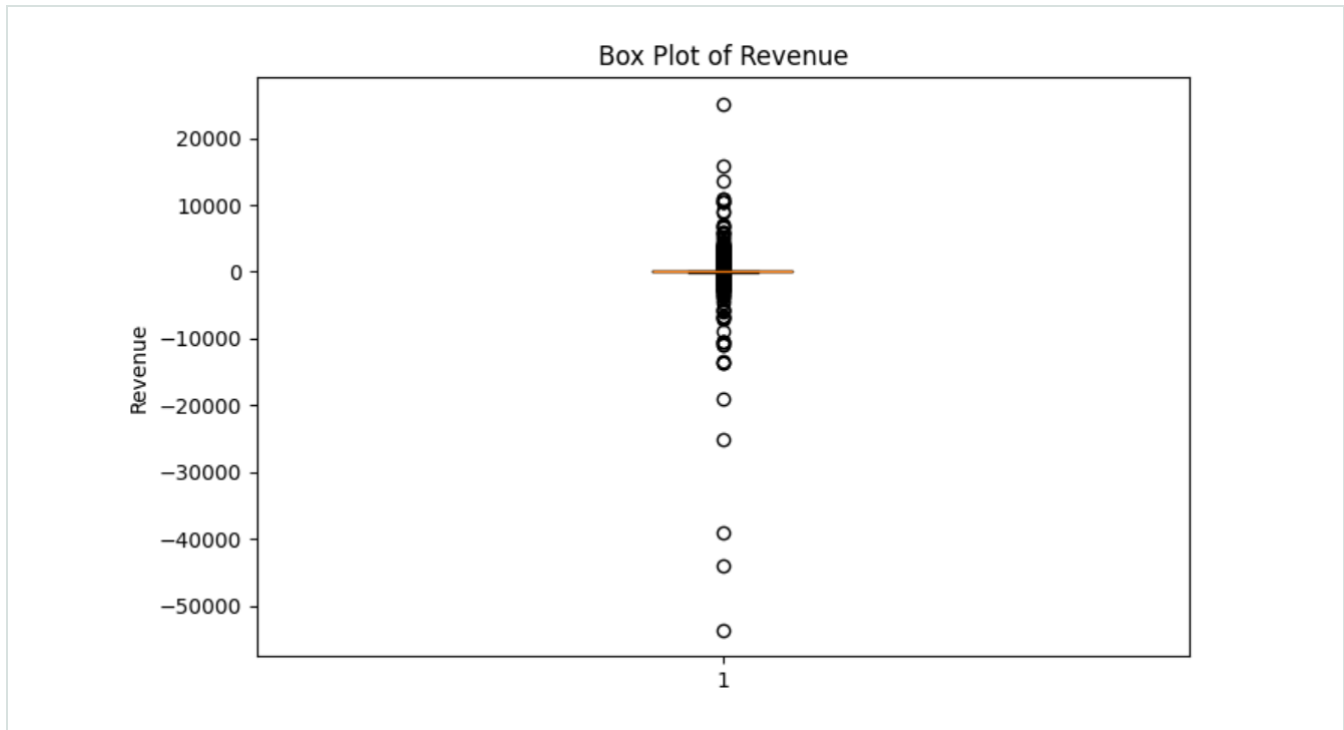


Figure 5.9 — Box plot: Revenue

Box plots revealed several extreme outliers in Quantity, Price, and Revenue. These outliers may represent bulk purchases, product returns, or high-value transactions. No outliers were removed, as the objective of this task was observation only.

6 Business Insights

- Around 20.54% of the Customer ID values are missing, making customer-level analysis incomplete.
- The dataset contains 6,865 duplicate rows, indicating data quality issues that should be handled before predictive modeling.
- The United Kingdom contributes the highest revenue, making it the retailer's most important market.
- November records the highest monthly revenue, suggesting seasonal shopping behavior.
- Products with the highest sales quantity are not always the products generating the highest revenue.
- Price has a stronger impact on revenue than quantity sold, as shown by the correlation analysis.
- The dataset contains several extreme outliers, indicating unusual purchasing behavior such as bulk orders or returns.

7 Conclusion

The Exploratory Data Analysis provided valuable insights into the Online Retail II dataset. The analysis identified important data quality issues, including missing customer information and duplicate records.

Product-level analysis highlighted the retailer's best-performing products, while country-wise analysis showed that the United Kingdom dominates total revenue.

The monthly revenue trend demonstrated seasonal variations, with the highest sales occurring during November. Correlation analysis indicated that product price contributes more significantly to revenue than sales quantity. Finally, the presence of numerous outliers suggests the existence of bulk purchases and return transactions that should be carefully considered during future data preprocessing.

Overall, this analysis successfully fulfilled all assignment requirements and established a strong foundation for future machine learning tasks, including customer segmentation, sales forecasting, and recommendation system development.